

SOC INCIDENT REPORT

Project 1: Brute-Force Authentication Investigation

Analyst: Daniela Robertson

1. Incident Summary

Incident Type	Potential Brute-Force Authentication Activity
Date Observed	September 18, 2026
Severity	High
Status	Under Investigation

Executive Summary

On September 18, 2026, beginning at 9:55:00 AM, the jlee account experienced 18 failed authentication attempts originating from source IP 185.77.12.44. The failed attempts were followed by a successful authentication from the same source IP at 10:02:10 AM. This source IP differed from previously observed authentication activity for jlee and other users in the provided log, whose normal activity originated from the 10.10.20.x range. The repeated authentication failures followed by a successful login from a different source IP are consistent with potential brute-force activity and warrant further investigation.

2. Evidence

Evidence	Finding
Suspicious username	jlee
Suspicious source IP	185.77.12.44
Failed attempts	18 failed authentication attempts
Successful authentication	jlee successfully authenticated from 185.77.12.44 at 10:02:10 AM
Relevant time window	9:55:00 AM-10:02:10 AM on September 18, 2026
Previous observed source IP	jlee previously authenticated successfully from 10.10.20.15

3. Timeline

Timestamp	Event	User	Source IP	Result
9:55:00 AM	First suspicious authentication attempt	jlee	185.77.12.44	Failed
9:55:00 AM-10:01:14 AM	18 authentication attempts observed	jlee	185.77.12.44	Failed
10:01:14 AM	Final failed authentication attempt	jlee	185.77.12.44	Failed
10:02:10 AM	Authentication attempt	jlee	185.77.12.44	Successful

4. Analysis

Based on the available evidence, the authentication activity is consistent with possible brute-force activity. The jlee account experienced 18 failed authentication attempts within a short time frame, all originating from

source IP 185.77.12.44. This source IP differed from the IP address previously associated with successful jlee authentication activity. Following the 18 failed attempts, a successful authentication from 185.77.12.44 occurred at 10:02:10 AM, 56 seconds after the final failed attempt. Although this pattern is suspicious, the available authentication logs alone do not confirm that the account was compromised or that the successful login was unauthorized.

5. MITRE ATT&CK Mapping

Technique	Brute Force
Technique ID	T1110
Reasoning	The jlee account experienced 18 consecutive failed authentication attempts from source IP 185.77.12.44, followed by a successful authentication from the same IP. This pattern is consistent with potential brute-force activity.

6. Recommended Response

1. Temporarily disable the jlee account while the suspicious authentication activity is investigated.
2. Initiate a password reset to invalidate potentially compromised credentials.
3. Verify with the account owner whether the successful authentication from 185.77.12.44 was authorized.
4. Review subsequent account and system activity for evidence of unauthorized access and preserve relevant logs.
5. If MFA is not already enabled, require multifactor authentication to strengthen account security.

7. Analyst Conclusion

Based on my investigation, I assess that the suspicious authentication activity involving the jlee account demonstrates a pattern consistent with possible brute-force activity. This assessment is based on 18 failed authentication attempts from an unusual source IP, followed by a successful authentication from the same IP. However, the available evidence does not confirm that the account was compromised or that the successful authentication resulted from a brute-force attack. Further investigation should review the activity performed by the jlee account following the successful authentication and determine whether the account owner recognizes the login.

Training note: This incident report is based on a synthetic authentication dataset created for a cybersecurity portfolio project. No real users or systems are involved.